

The Embedded-Chain RGA

A tombstone-free replicated list whose coordinates are the birth chains,
entropy-optimally encoded — design, proofs, validation, and related work

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working names in the artifact: `embed-tree` (unary), `embed-code` (delta-coded)

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Abstract

The embedded-chain RGA is a replicated list (a sequence MRDT for three-way merges with an LCA) whose entire state is its live nodes: no tombstones, no ledger, no event log. Each node carries one parent-relative dyadic interval; the insert *mint* embeds the timestamp delta to the anchor in an order-preserving prefix-free code, which makes concurrent same-position mints provably non-colliding — *ties cannot exist* — so the merge contains no repair and never decides an order. Every node’s absolute interval is a birth constant; display order coincides with lexicographic order on birth chains; the state is a canonical function of the event set. Deletion is an exact isometric fold, so a dead node’s interval encloses its descendants forever: the dead survive only as $\Theta(\log \delta)$ coordinate bits inside living positions — the information-theoretic minimum, paid only at genuine races. The design is machine-validated: clean on the L1–L25 anomaly battery (except backward-run interleaving L19, the published RGA’s own anomaly) and on 420+ randomized version-DAG executions, and **lockstep read-equal with the published tombstoned RGA at every step of 120 randomized executions**. Insert operations carry the anchor’s coordinate prefix — consulted by no reachable execution, kept because the proof of `rc = Either` commutation requires totality of reordered application; the prefix is ghost state in the strict, machine-checkable sense. We give the component-by-component definition, the four theorems with proofs, the validation record, a detailed comparison with Eg-walker, related work, and the Lean mechanization plan on the proved RGA scaffolding.

1 The datatype, component by component

Timestamps $t \in \mathbb{N}^+$ are Lamport stamps, globally unique, and double as element identities. Causality gives: the anchor of every insert carries a smaller timestamp than the insert itself.

Type. A state is a finite map over *live nodes only*:

$$\sigma : t \mapsto (\text{par} \in \mathbb{N}, (lo, hi) \subset (0, 1), \text{char}),$$

where (lo, hi) is a dyadic-rational interval *relative to the parent’s interval*, and $\text{par} = 0$ denotes the root sentinel (which owns $(0, 1)$ and is never stored). A node’s *absolute* interval is the affine composition of the intervals along its parent chain. Semantically the coordinates are rationals; the natural carrier is a bit string per node (§5).

init. $\sigma_0 = \emptyset$.

The mint. For $\delta \geq 1$ let L be the bit-length of δ and

$$C(\delta) = \underbrace{1 \cdots 1}_{L-1} 0 \text{ } (\delta \text{ with its leading bit removed}) \quad (|C(\delta)| = 2L - 1),$$

an order-preserving prefix-free code. Writing $b = 1 \cdot C(\delta)$ (a leading 1 keeps coordinates positive) and $w = 2^{-|b|-2}$, the mint is the second quarter of b 's dyadic cell:

$$M(\delta) = (0.b + w, 0.b + 2w) \subset [0.b, 0.b + 4w).$$

The quarter placement leaves a gap below and headroom above inside the cell, and the open space $(0.b + 2w, 0.b + 4w)$ is never minted — it exists so that the interval has room to *contain* the node's own subtree without touching the next cell.

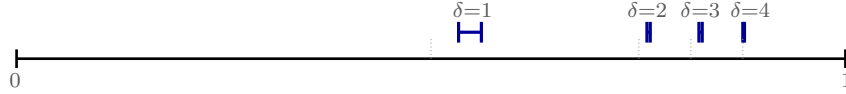


Figure 1: Mints $M(\delta)$ for $\delta = 1..4$ inside the parent's unit interval (cell boundaries dotted). Larger deltas sit strictly higher; distinct deltas are disjoint with gaps; each mint is a quarter of its cell, leaving room for the node's own subtree.

do (the operations). Two operations, with $rc = \text{Either everywhere (§3)}$:

- $\text{ins}(x \leftarrow a, c, \pi)$ — insert element x (its timestamp) with character c after anchor a ($a = 0$ for the front). The payload π is the anchor's coordinate prefix, carried *for the proof alone* (§3). Application on a state where a is live (every reachable application, by honesty):

$$\sigma[x] := (a, M(x - a), c), \quad \pi \text{ unread.}$$

No head-tracking, no reading of siblings: the mint depends only on the two timestamps.

- $\text{del}(d)$ — remove d 's record and *isometrically fold* it into its children: each child c of d is re-parented to d 's parent with d 's interval composed in, $c.\text{frac} := d.\text{frac} \circ c.\text{frac}$. This is an exact affine substitution: every survivor's absolute interval is unchanged. If d is absent, no-op (concurrent double delete).

read. Depth-first from the root, children in descending order of lo . No tie rule is needed (Corollary 1).

merge. $\text{merge}(L, A, B)$, L the LCA:

1. *Survival* (OR-set): live iff in all three, or born in exactly one branch.
2. *Refold*: for each survivor, compute its absolute interval by folding its record to the root inside the input that holds it (a record's stored parent is always live in its own input, so the chain stays in one input; by Theorem 2 the choice of input cannot matter — asserted at runtime, never fired); re-parent to the nearest *surviving* ancestor; re-relativize.

There is no step 3. No overlap repair exists because overlaps cannot occur (Theorem 3); the merge never compares timestamps and never decides an order.

Inv / applicable / honesty. The honesty contract is the proved RGA's: an insert is *generated* at a replica where its anchor is live (you insert after a character you can see), and ins/del are *applied* on states where they are applicable (anchor live; delete of a live or already-dead node). On every reachable state, application never consults π . The representation invariant (Theorem 4) doubles as Inv.

$\approx$. Equality. The state is a canonical function of the event set (Theorem 4), so no quotient is needed — convergence is literal state equality.

Worked example. 6 and 10 concurrently insert at the front ($\delta = 6, 10$): disjoint mints, 10's higher — display $[10, 6]$, no repair, at every replica and merge order. Type 22, 16 under 6; delete 6: the fold composes 6's interval into 22's and 16's — they remain inside the region $M(6)$, below $M(10)$, forever. A third party concurrently inserts 8 at the front: $M(8)$ lands between $M(6)$ and $M(10)$ by Theorem 1 — display $[10, 8, 22, 16]$ whether 8 arrives before or after 6's death (machine-checked as the *credential countermodel*, the shape that kills every timestamp-oblivious variant, §4).

2 Theorems

Theorem 1 (mint disjointness and order). *For $\delta \neq \delta'$, $M(\delta)$ and $M(\delta')$ are disjoint with a gap; $\delta < \delta' \Rightarrow M(\delta)$ lies entirely below $M(\delta')$.*

Proof. C is prefix-free: codes of lengths $2L-1 < 2M-1$ differ at position L (the shorter's header terminator 0 against the longer's header 1); equal-length codes differ in the payload. Prefix-free codes have pairwise-disjoint dyadic cells. C is monotone as a binary fraction: within a length class the payload is δ 's low bits in order; across classes the position- L bit decides in favor of the longer code. Disjoint cells ordered by their codes, and mints interior quarters of their cells, give both claims. $\square$

Theorem 2 (birth constancy). *Define $\alpha(u) = \varphi_{c_1} \circ \dots \circ \varphi_{c_k}(M(\delta_u))$, the affine composition of the mints along u 's birth chain $c_1 \dots c_k u$ (where φ_v maps $(0, 1)$ onto v 's mint interval, order-preservingly, and δ_v is v 's delta to its birth parent). Then in every reachable state, every live node's absolute interval equals $\alpha(u)$.*

Proof. By induction over transitions. ins establishes it by definition. del's fold replaces (par, frac) pairs by their exact composition — the function α is unchanged. The merge recomputes precisely $\alpha(u)$ (folding a record chain to the root) and re-relativizes against another α ; ties cannot exist, so no step of any transition ever rewrites an interval by anything but exact composition. $\square$

Theorem 3 (no overlap, globally and permanently). *For distinct live u, v : either one's birth chain is a prefix of the other's (α nested — and the two are never read-time siblings), or $\alpha(u) \cap \alpha(v) = \emptyset$.*

Proof. If neither chain is a prefix of the other, they first differ at some level, on distinct deltas relative to the *same* birth ancestor; Theorem 1 separates them there, and affine images of disjoint intervals are disjoint. If u lies on v 's birth chain and both are live, v 's nearest live ancestor is u or deeper — so they never share a current parent, and sibling groups are pairwise disjoint. Permanence is Theorem 2. $\square$

Theorem 4 (canonicity). *The state is a function of the event set: $\sigma = \{u \mapsto (\text{lp}(u), \alpha(u) \text{ relative to } \alpha(\text{lp}(u)), \text{char}_u)\}$ over survivors u , where $\text{lp}(u)$ is the nearest live birth-ancestor.*

Proof. Induction: `ins` adds exactly this record (its anchor is live at application, by honesty); the fold rewrites a dead parent’s children to exactly this form (exactness); the merge computes exactly this form, and survivorship is a function of the event set (OR-set). $\square$

Corollary 1 (no tie rule). *Sibling lo-values never tie; the read’s id tiebreak is dead code. Machine check: pairwise disjointness asserted at every read of 320 randomized executions — clean.*

Corollary 2 (`display` $\equiv$ `chain-lex`). *Display order is the lexicographic order on birth chains (per level: larger delta first — among same-parent siblings, exactly newest first): survivors rank by their own timestamps, folded heirs by their dead founder’s. Machine check over 19,531 multi-member sibling groups: own-timestamp order violated 3,481 times (exactly the post-fold heir groups — forcing own-ts there is the refuted dissolution behavior), **chain-lex violated 0 times**. Equivalently: the canonical order is the RGA order of the birth tree restricted to survivors.*

3 The insert prefix: for the proof, and the proof alone

The proved tombstone-free RGA’s *why-the-path-matters* analysis applies verbatim, and this design inherits its discipline.

Why rc must be Either. In the Sal framework, any oriented conflict-resolution entry for an insert/delete pair awakens the machine-refuted `cond_comm_base` obligation: ordering `ins( $x \leftarrow a$ )` against `del( $a$ )` (in either direction) is undischARGEABLE as a conflict entry, because rehoming is non-transitive across chains of deletions. So `rc = Either`: the proof must show the two *commute*, in both orders, on all applicable states.

The commutation, and where the prefix earns its keep. Order 1: `ins( $x \leftarrow a$ )` then `del( $a$ )` — x is born under a , then the fold re-homes x to a ’s parent with composed coordinates; by exactness, x ’s absolute interval is $\alpha(a) \circ M(x - a)$. Order 2: `del( $a$ )` then `ins( $x \leftarrow a$ )` — a is gone; application must still be *total* and land x at the same α . This is what the carried prefix π (the anchor’s coordinate, equivalently its chain of codes) provides: apply computes $\alpha(x) = \pi \circ M(x - a)$ and attaches x to the nearest live ancestor. The two orders produce *equal* states — commutation to equality, not merely up to $\approx$, courtesy of Theorem 4.

Ghost, in the strict sense. On every reachable state the anchor is live at application (honesty), and the mint reads only the two timestamps: π is never consulted. The Lean artifact should carry the analogue of the proved RGA’s `ins_path_free`: a machine-checked lemma that the transition function restricted to reachable states is invariant under erasure of π . An implementation may therefore drop π from the wire entirely; the *proof* may not, because the verification conditions quantify over reorderings that visit unreachable states, and there application without π is partial. The prefix costs $O(\sum \log \delta)$ bits — the same entropy as the coordinate itself — and exists at the proof layer alone.

4 Validation

All numbers reproduce from `whiteboard/litmus/` (`embed_tree.py`; harnesses `litmus.py`, `pbt.py`).

check	verdict
litmus battery L1–L25 (incl. every countermodel that killed 16 prior designs: ceiling escapes, tie inheritance, three-branch topology, frame mixing, fold-verdict inheritance)	clean, except one-sided L19 (backward-run interleaving — the published RGA’s own anomaly)
credential countermodel (tie heirs vs mid-ts third party, death-before vs death-after topologies)	converges, no flips — = RGA _† ’s reads
subordination-escape and retroactive-subordination CEs	survive
randomized version-DAG PBT (FLIP / CONV / LIVE / DUP)	120/120 and 300/300 clean
per-read sibling-disjointness assertion sweep	320/320 clean
chain-lex instrumentation (19,531 sibling groups)	0 violations
lockstep vs the published tombstoned RGA	read-equal at every apply/merge/read, 120/120
L1 delete-reorder vs the proved flat tombstone-free RGA	embed [2, 3] ✓ vs flat [3, 2] ×

The lockstep row is the identification: *the embedded-chain RGA is observationally the published tombstoned RGA with the tombstone set compressed into coordinate entropy* — machine-checked on 120 executions; the read-for-read theorem is the mechanization target, not yet a theorem. The L1 row is the separation from the repo’s *proved* flat tombstone-free RGA: that design’s climb re-sorts a rehomed node by its own timestamp (the $[b, a, c] \rightarrow [c, b]$ anomaly), i.e. its sort key is the birth chain *truncated at each rehoming*; this design keeps the chain whole.

5 Cost and representation

The entropy law. A coordinate is the concatenation of the codes along the birth chain: $\Theta(\sum_i \log \delta_i)$ bits — the entropy of the chain. Sequential typing ($\delta = 1$ at every level) costs ~ 4 bits per level; a race with timestamp gap g costs $2 \log_2 g + O(1)$ bits. *The state pays for the magnitude of actual concurrency, and for nothing else.* Measured (1000 elements):

scenario	unary mint	delta-coded mint	quarter carve (ts-oblivious, refuted)
chain then delete-all ($\delta=1$)	2^{502501} (~ 61 KB)	2^{4001} (~ 0.5 KB)	2^{2001}
1000 root siblings ($\delta=t$)	$\max 2^{1003}$, 125 KB total	max 2^{23}, 5 KB total	— (ties)

The carrier. The natural representation is a **parent-relative bit string per node**: the fold is concatenation, comparison is lexicographic — $O(1)$ -ish with structure sharing; the rationals are the semantics of the strings, kept for the proofs. IEEE 754 binary64 is machine-refuted as the carrier: mints degenerate at $t = 52$, depths underflow at 44, and rounding voids Theorem 2 (6/120 PBT executions die). The obstruction is a pigeonhole — chains carry $\Theta(\sum \log \delta)$ bits, a fixed-width word holds 64 — but binary64 is sound as a *comparison cache* with exact fallback on float equality.

What remains open. Dead chains concatenated into survivors’ strings are the conservation law’s irreducible content; reclaiming them under *causal stability* (renumber once every replica has seen the delete — the only condition under which rewriting geometry is safe, since no verdict involving the dead can be contested again) is the one open cost item, and it is orthogonal to correctness.

6 Comparison with Eg-walker

Eg-walker (Gentle & Kleppmann, EuroSys 2025) is the closest system in spirit, and the two designs can be read as the *operational* and *denotational* realizations of the same thesis — **pay for**

concurrency, not for history:

	Eg-walker	embedded-chain RGA
model	op-based; replicas exchange events	MRDT: three-way merge with an LCA
durable state	full event graph on disk , deletes included, retained while concurrent ops may arrive; document state metadata-free	live nodes only ; no event log anywhere; dead survive as $\Theta(\log \delta)$ bits inside live coordinates
merge mechanism	<i>replay</i> : walk the event graph, rebuild a transient CRDT structure (tombstones reappear transiently), discard at causally-linear points	<i>refold</i> : $O(\text{survivors})$ coordinate recomputation; no replay, no transient structure, no order decisions
sequential-editing cost	$\approx$ zero (plain edits; no metadata)	$\approx$ zero order metadata (~ 4 bits/level; plain append-like behavior)
concurrency cost	replay time proportional to the concurrent region, at every affected merge	coordinate bits proportional to $\log(\text{ts gap})$, once, in space
ordering semantics	Yjs-variant (FugueMax-adjacent); maximal non-interleaving <i>conjectured</i> , not proved; two-sided (no L19 analogue)	$\equiv$ published RGA (machine lockstep); one-sided — L19 backward-run interleaving inherited, two-sided extension future work
correctness artifact	informal proof (Appendix C) + property testing	L1–L25 battery + DAG PBT + lockstep vs RGA ₊ ; Lean mechanization as the deliverable — the design was chosen so the merge proves itself (no order decisions to verify)
op payload	integer origin positions (compact, human-meaningful)	two timestamps + ghost prefix (droppable on the wire; §3)

Three sentences of honest positioning. (i) Eg-walker achieves its clean document state by keeping the entire history and re-deriving order on demand; the embedded-chain RGA achieves it by making order derivation unnecessary — the coordinates *are* the derivation, precomputed at mint time and exact forever. (ii) Eg-walker’s ordering target is stronger where L19 is concerned (Fugue-style two-sidedness removes backward interleaving; this design inherits RGA’s one-sided anomaly — by construction, since it is read-equal to RGA). (iii) The verification gap runs the other way: Eg-walker’s replay-transform-clear loop is algorithmically sophisticated and only informally proved; here the merge performs no arbitration at all, the four theorems are one-paragraph proofs, and the mechanization inherits a proved scaffolding (§8). In the MRDT setting there is also a model fit: Eg-walker must reconstruct merge context by replay because the op-based model gives it none; the MRDT model hands the merge its LCA, which is exactly the context replay rebuilds.

7 Related work

The verified landscape sweep lives in `whiteboard/litmus/related-work.md` (per-claim verdicts, primary sources); this section positions the datatype.

Retention taxonomy. What nearby systems keep for the dead: tombstone records (RGA with cemetery under causal stability; WOOT; Yjs’s per-deletion GC structs, kept forever; Fugue — “we cannot remove a deleted element’s node entirely”; Peritext — “the positions must be remembered”); the full history (Eg-walker’s event graph, Chronofold/causal trees where the log *is* the text); consensus-gated compaction (Treedoc’s flatten requires Paxos-commit); or **dead identifiers inside live position keys** (Logoot/LSEQ, Weidner’s position-strings/list-positions — the Fugue id space). The embedded-chain RGA is in the last class *by information content* — dead deltas persist as code bits inside survivors’ coordinates, at live-reachable scope — but with three deltas to its classmates: the retention is *entropy-optimal and delta-coded* (a sequential history costs a constant per element where Logoot-family keys grow with allocation policy), the *tree with isometric fold* gives forward non-interleaving and delete-order preservation (position-key designs interleave — the PaPoC’19 anomaly — and Figma-style fractional indexing is unsound under concurrency by its own account), and the semantics is pinned to the best-studied sequence CRDT by a machine-checked lockstep rather than being a new point in anomaly space.

The impossibility backdrop. Attiya et al. (PODC 2016) prove $\Omega(D)$ metadata for push-based protocols meeting even the weak list spec — the published form of “ordering requires remembering the dead.” This design does not escape the bound; it *relocates and minimizes* the memory: $\Theta(\log \delta)$ bits per dead-chain element inside live coordinates (plus the MRDT version store, which the model provides). The campaign’s sharper, machine-witnessed form is the *conservation law*: the verdict information survives every encoding change — stamp, spine, tombstone, or denominator bits — and ties force separate *metadata* only under timestamp-oblivious minting; the timestamp-faithful mint carries the same bits inside the coordinate. Eight timestamp-oblivious mutation variants each have a named machine-checked countermodel (fold-verdict erasure; repair non-locality); the credential countermodel separates this design from all of them.

Verified datatypes. Isabelle RGA (Gomes et al. OOPSLA’17, op-based convergence), OpSets (list semantics specified by a total order of identified ops — arbitration-by-identity as specification), VeriFx (SMT, convergence properties), MET (TLA+, list CRDT bugs). In the MRDT line (Kaki et al. OOPSLA’19; Peepul PLDI’22; RA-lin OOPSLA’25 — this framework), no verified sequence with sequence-CRDT-grade ordering guarantees exists; the proved flat tombstone-free RGA in this repo is own-fold RA-linearizable but fails the sequential delete-order spec (the spec-limit lesson). The embedded-chain RGA targets: the first verified tombstone-free sequence whose *intent* theorems (delete-order preservation, forward non-interleaving, RGA read-equivalence) are part of the statement, not excluded by it.

Treedoc’s lesson, inverted. Treedoc re-balances geometry and must pay consensus for it; the delta-tree arc’s principle (*reprojection without consensus is safe iff geometry has been demoted to a rendering*) is here taken to its fixed point: geometry is never reprojected at all — it is immutable, so it may safely be the only thing.

8 Mechanization plan

Rides the proved tombstone-free RGA’s scaffolding (Sa1/MRDTs/RGA): ConditionedMRDTSig; rc = Either everywhere; prefix-carrying insert with the ghost-erasure lemma (§3); the honesty contract (born-accuracy + applicability) and the HonestReach/GenHonest metatheorems unchanged. What is new is only the read/state layer, and it is *smaller* than the flat RGA’s:

1. **Spec layer:** lists ordered by chain-lex (Corollary 2); convergence = determinism of a function of the event set (Theorem 4 makes $\approx$ equality — no quotient); pairwise display stability from immutability of α .
2. **Refinement:** the coordinate encoding is an order-embedding of chain-lex — Theorems 1 and 3, each a one-paragraph induction; fold exactness is algebra on affine maps.
3. **VC discharge:** the ins/del commutation is *to equality* (§3); ins/ins commute by disjointness; del/del by set algebra. No merge case analysis exists because the merge makes no decisions.
4. **Capstone:** own-fold RA-linearizability (the framework route), *plus* the intent theorems the flat RGA provably lacks — delete-order preservation (L1 as a SPOT theorem), forward non-interleaving — *plus* the equivalence target $\text{read} \circ \text{embed} = \text{read} \circ \text{RGA}_\dagger$ on honest executions, which converts the design into a theorem *about* the published RGA: it admits a strictly-live compaction preserving every read.

Until the Lean theorems land, every claim above is scoped as machine-checked on the stated executions, not proved.

Provenance. `whiteboard/litmus/embed_tree.py` (the design, the timestamp-oblivious control, the credential countermodel, the IEEE 754 measurements; `python3 embed_tree.py / fp / code`); `litmus.py`, `pbt.py` (battery and DAG harness); `contest_tree.py` (lockstep harness, CE regressions); `whiteboard/retention-countermodel.pdf` (the retention arc that forced this design); `whiteboard/delta-tree-design.pdf` (the ledger-based sibling design and the invariants I1–I3); `whiteboard/litmus/related-work.md` (verified source list: Roh et al. JPDC’11; Attiya et al. PODC’16; Kleppmann et al. PaPoC’19; Weidner & Kleppmann TPDS’25 (Fugue); Gentle & Kleppmann EuroSys’25 (Eg-walker, arXiv:2409.14252); Preguiça et al. ICDCS’09 (Treedoc); Grishchenko & Patrakeev PaPoC’20 (Chronofold); Litt et al. CSCW’22 (Peritext); Gomes et al. OOPSLA’17; OpSets 2018; Kaki et al. OOPSLA’19; Peepul PLDI’22; RA-lin OOPSLA’25; VeriFx ECOOP’23; Figma/Wallace fractional indexing; Weidner position-strings).